

1. Probability and conditioning

1. Axioms and Bayes

- $P(A) \geq 0, P(\Omega) = 1$.
- If A_i are disjoint, $P(\bigcup_i A_i) = \sum_i P(A_i)$.
- $P(A | B) = \frac{P(A \cap B)}{P(B)}$ for $P(B) > 0$.
- $P(A \cap B) = P(A | B)P(B) = P(B | A)P(A)$.
- Partition rule: $P(A) = \sum_i P(A | B_i)P(B_i)$.
- Bayes: $P(H | E) = P(H) \frac{P(E | H)}{P(E)}$.

Pitfall. Conditional probability is a renormalized probability on B , not $P(A \cap B)$.

2. Random variables

- A random variable is a function $X : \Omega \rightarrow \mathbb{R}$.
- Induced law: $P_{X(S)} = P(X \in S)$.
- Joint PMF: $p_{X,Y}(x, y) = P(X = x, Y = y)$.
- Marginals: $p_{X(x)} = \sum_y p_{X,Y}(x, y), p_{Y(y)} = \sum_x p_{X,Y}(x, y)$.
- Conditional PMF: $p_{X|Y}(x|y) = p_{X,Y} \frac{x,y}{p_{Y(y)}}$ when $p_{Y(y)} > 0$.
- Independence: $X \perp Y$ iff $p_{X,Y}(x, y) = p_{X(x)}p_{Y(y)}$ for all x, y .

Pitfall. Probability lives on events, not on raw outcomes; a random variable is the map from outcomes to values.

2. PMF / PDF / CDF

1. Discrete laws

- Bernoulli(p): $P(X = 1) = p, P(X = 0) = 1 - p$.
- Binomial(n, p): $p_{X(k)} = \binom{n}{k} p^k (1 - p)^{n-k}, k = 0, \dots, n$.
- Geometric(p): $p_{X(k)} = (1 - p)^{k-1} p, k = 1, 2, \dots$.
- Poisson(λ): $p_{X(k)} = e^{-\lambda} \frac{\lambda^k}{k!}, k = 0, 1, \dots$.
- Discrete uniform on $\{a, \dots, b\}$: $p_{X(i)} = \frac{1}{b-a+1}$.

Pitfall. Many binary observations are not automatically binomial; check the assumptions first.

2. Continuous laws

- PDF: $f_{X(x)} \geq 0, \int f_{X(x)} dx = 1$.
- $P(a \leq X \leq b) = \int_a^b f_{X(x)} dx$.
- CDF: $F_X(x) = P(X \leq x)$.
- If differentiable, $f_X(x) = dF_X \frac{x}{dx}$ and $F_X(x) = \int_{-\infty}^x f_X(t) dt$.
- Continuous: $P(X = c) = 0$ for any fixed c .
- Uniform(a, b): $f_{X(x)} = \frac{1}{b-a}$ on $[a, b]$.
- Exponential(λ): $f_{X(x)} = \lambda e^{-\lambda x}$ for $x \geq 0$.
- Gaussian(μ, σ^2): $f_{X(x)} = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.

Pitfall. A PDF is a density, not a probability. Only integrals over intervals give probabilities.

3. Expectation and variance

1. Core identities

- Discrete: $E[X] = \sum_x x p_{X(x)}$.

- Continuous: $E[X] = \int x f_{X(x)} dx$.
- $\text{Var}(X) = E[X^2] - (E[X])^2$.
- $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$.
- $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$.
- $E[\mathbf{1}_A] = P(A)$ and $\text{Var}(\mathbf{1}_A) = P(A)(1 - P(A))$.

2. Useful rules

- Linearity: $E[aX + bY] = aE[X] + bE[Y]$.
 - If $X \perp Y$, then $E[XY] = E[X]E[Y]$ and $\text{Cov}(X, Y) = 0$.
 - Total expectation: $E[X] = E[E[X|Y]]$.
 - Total variance: $\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$.
- Pitfall.** $\text{Var}(X + Y)$ collapses to $\text{Var}(X) + \text{Var}(Y)$ only under independence.

4. Transforms and information

1. Transformations

- If $Y = g(X)$, then $F_{Y(y)} = P(g(X) \leq y)$.
- Inverse transform: $U \sim \text{Uniform}(0, 1)$ and $X = F_X^{-1}(U)$ gives CDF F_X .
- Probability integral transform: if F_X is continuous and strictly increasing, then $F_X(X) \sim \text{Uniform}(0, 1)$.
- Discrete convolution: $p_{Z(z)} = \sum_y p_{X(z-y)} p_{Y(y)}$ for $Z = X + Y$ and $X \perp Y$.

- Continuous convolution: $f_{Z(z)} = \int f_{X(z-y)} f_{Y(y)} dy$.

2. Entropy and KL

- $H(P) = -\sum_x P(x) \log P(x)$.
- $D_{\text{KL}}(P \parallel Q) = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) \geq 0$.
- Cross entropy: $H(P, Q) = H(P) + D_{\text{KL}}(P \parallel Q)$.

Pitfall. KL is mismatch cost: it is zero only when $P = Q$, and it is not symmetric.

5. Model criticism and mixtures

1. Gaussian mixtures

- Mixture model: $p(x) = \sum_j w_j p_{j(x)}$ with $w_j \geq 0$ and $\sum_j w_j = 1$.
- Mean: $E[X] = \sum_j w_j E_{j[X]}$.
- Variance: $\text{Var}(X) = \sum_j w_j (\text{Var}_{j(X)} + (\mu_j - \mu)^2)$.

2. Model checking

- Compare observed frequencies to the model's predicted probabilities.
 - Look for systematic deviation, not just one-off outliers.
 - Check whether assumptions match the data-generating process.
- Pitfall.** A good fit on one summary statistic does not mean the whole model is correct.